A zebrafish model of ATP1A3-related disorders links telencephalic dysregulation with sensorimotor coordination and movement control

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Abstract

Mutations in ATP1A3, encoding the neuron-enriched Na^+/K^+-ATPase $\alpha 3$ subunit, cause a spectrum of severe neurological disorders that commonly feature episodic and stress-triggered motor symptoms, including dystonia, ataxia, and hemiplegic attacks. Despite strong evidence that impaired ionic homeostasis underlies these conditions, the specific brain regions, circuit mechanisms, and physiological dynamics most vulnerable to ATP1A3 loss remain unclear. Here we generated a larval zebrafish model of ATP1A3-related disease by creating a frameshift knockout of the zebrafish *atp1a3a* gene and characterized resulting behavioural and neurophysiological abnormalities. Mutant larvae exhibit robust disruptions in both spontaneous and stimulus-evoked behaviour, including abnormal postures and dyscoordination consistent with key clinical features. To identify affected circuit substrates, we performed whole-brain functional imaging and found a reproducible forebrain network exhibiting pronounced dysregulation, including reduced stimulus tuning and episodes of broadly coordinated activity followed by sustained silencing. Together, these findings support a model in which forebrain circuits are particularly sensitive to *atp1a3a* loss and establish larval zebrafish as a tractable platform for dissecting molecular, cellular, and circuit-level mechanisms that may drive ATP1A3-related neurological symptoms.

Introduction

Mutations in the ATP1A3 gene, encoding the Na^+/K^+-ATPase $\alpha 3$ subunit, are responsible for a spectrum of rare neurological disorders collectively termed ATP1A3-related disorders. The Na^+/K^+-ATPase is an ATP-driven membrane pump that exports three Na^+ ions and imports two K^+ ions per cycle to maintain the ionic gradients that set neuronal excitability, and the neuron-enriched $\alpha 3$ subunit is particularly important for rapidly restoring Na^+ homeostasis after high-frequency firing, providing "reserve" pumping capacity during periods of high demand (Holm and Lykke-Hartmann, 2016). ATP1A3-related disorders include Alternating Hemiplegia of Childhood (AHC), Rapid-Onset Dystonia-Parkinsonism (RDP), Cerebellar Ataxia, Areflexia, Pes Cavus, Optic

Atrophy, Sensorineural Hearing Loss (CAPOS), Relapsing febrile encephalopathies (RECA / FIPWE), and certain forms of early-onset disorders such as Developmental/Early-Onset Epileptic Encephalopathy (DEE), sometimes accompanied by polymicrogyria (PMG) (Salles et al., 2021; Li et al., 2022; Vezyroglou et al., 2022).

While distinct diagnoses, these diseases share the common genetic basis of the ATP1A3 gene, and commonly exhibit both strongly disordered movements patterns (dystonia, ataxia, etc), as well as other cognitive impairments, often affecting memory, processing speed, and executive function, as well as psychiatric manifestations such as mood disorders and anxiety (Pittock et al., 2000; Sweney et al., 2009). While dysregulation of ionic gradients and neuronal excitability due to the dysfunctional pump is very likely the root cause of most of these disease symptoms, the variability among disorders associated with specific genetic alleles of ATP1A3, and the variability among patients, suggests a relatively complex and somehow specified pathology that may have differing involvement of different neurological systems (Li et al., 2022). Interestingly, symptoms are often triggered or exacerbated by stress, fever, or exertion, and come in waves/attacks, indicating that the underlying disease condition is somehow transitory and correctable, or varies in severity across time (Salles et al., 2021). The brain areas, circuit- and cellular-level mechanisms that are vulnerable to ATP1A3-loss and underlie disease states, as well as these sources of symptomatic transience, are not clear.

Dystonia is a common feature of many ATP1A3-related disorders, characterized by sustained or intermittent muscle contractions that cause abnormal postures or twisting movements (Albanese et al., 2013). In RDP, dystonia is a central symptom that often has an abrupt onset in adolescence or early adulthood (Haq et al., 2019). The neurological basis of dystonia is thought to be centered in the basal ganglia (BG), which are important for refining motor programs, such as selecting appropriate actions and suppressing unwanted movements (Mink, 2018). Disruptions of these systems are thought to lead to excessive facilitation of motor output in dystonia and a reduced ability to suppress movements (Mink, 2003). However, dystonia has been increasingly understood as a disorder of distributed brain networks. Functional imaging studies reveal abnormal connectivity between the BG, cerebellum, thalamus, and sensorimotor cortex in dystonia (Vo et al., 2023). Lesions of the cerebellum in animals can produce dystonic movements (Calderon et al., 2011), and patients with cerebellar disease may also develop dystonia (Alarcón et al., 2001). Ataxia, which is characterized by a loss of coordinated, accurate movements, often arises from cerebellar dysfunction (Kuo, 2019), and is also often observed in ATP1A3-related disorders, thus further implicating the cerebellum as a core locus of vulnerability in ATP1A3-related disease.

In addition to dystonia and ataxia, episodic hemiplegia is a defining feature of AHC, reflecting transient, lateralized failures of motor output that are inconsistent with focal structural lesions. Instead, hemiplegia points to dynamic dysfunction of distributed motor networks and impaired control of neuronal excitability. A leading mechanistic hypothesis is increased susceptibility to spreading depolarization (SD)—a slowly propagating wave of near-complete neuronal depolarization followed by prolonged synaptic silencing (Dreier, 2011). Because recovery from SD critically depends on Na^+/K^+-ATPase activity, ATP1A3 dysfunction is predicted to lower the threshold for SD initiation and prolong recovery (Hunanyan et al., 2014; Balestrino et al., 1999) This framework accounts for the defining features of AHC hemiplegic episodes, including abrupt onset, reversibility, sleep-related resolution, and alternating laterality across attacks. More broadly, an SD-based model could provide a unifying explanation for the coexistence of hemiplegia, dystonia, and ataxia in ATP1A3-related disorders, with differential motor symptoms arising from transient SD-based silencing of different brain regions or cell types.

Here we have generated a larval zebrafish model to better understand the role of ATP1A3 in neuronal circuits and to gain insights into potential cellular and circuit underpinnings of ATP1A3-related disorders. We generated a frameshift knockout of the zebrafish *atp1a3a*, and characterized its behavioural and neurophysiological abnormalities. Our mutants exhibit strong and widespread disruptions in both spontaneous and stimulus-evoked behaviour, some of which mirror clinical features of ATP1A3-related disorders, including aberrant postures and dyscoordination. Whole-brain functional imaging identified a common network in the forebrain that consistently shows dysregulated activity patterns including areas of the pallium, subpallium and habenula, characterized by a lack of stimulus-tuning and large bursts of coordinated activity among different groupings of neurons, followed by sustained silencing. Collectively, our data support a model in which forebrain circuits are particularly sensitive to ATP1A3-loss, and lay the foundation for using the larval zebrafish model to study both the molecular/cellular- and circuit-level mechanisms underlying forebrain-derived neurological symptoms in these diseases.

Results

Generation of a larval zebrafish model of ATP1A3-related disorders.

Due to the teleost genome duplication, zebrafish possess two genes that are orthologous to human ATP1A3: *atp1a3a* and *atp1a3b*. Zebrafish atp1a3a has 21 exons and shares 92% amino acid conservation with human ATP1A3, and a similarly broad gene expression in neurons throughout the central nervous system (Doğanli et al., 2013; Rajarao et al., 2001). The *atp1a3b* gene also shows very high conservation (91%), but its expression is highly localized to certain brain areas in the midbrain and hindbrain (Doğanli et al., 2013). For this project have focused on the more highly-conserved and more broadly-expressed atp1a3a gene. We generated a mutant line using CRISPR-Cas9, generating an 8 base pair deletion in exon 8 yielding a frameshift and premature stop codon in exon 9 (***Figure 1***A). qRT-PCR experiments show a strong loss of *atp1a3a* mRNA levels, indicating that the transcript is degraded through nonsense-mediated decay (***Figure 1***B). We tested for genetic compensation through paralog upregulation (Rossi et al., 2015), and indeed found that *atp1a3b* mRNA levels are significantly higher in *atp1a3a* mutants, indicating that atp1a3b overexpression may compensate for *atp1a3a* loss. Therefore, while we expect the *atp1a3a* allele to be completely non-functional, this may be partially compensated for by both the presence and upregulation of the *atp1a3b* paralog.

Previous experiments in zebrafish using morpholino-based knockdown of the zebrafish homologs of ATP1A3 (*atp1a3a* and *atp1a3b*) found brain ventricle dilation, as well as depolarization of the Rohon-beard neurons and motility deficits (Doğanlı et al., 2013). However, mutants for these genes did not exhibit these morphological abnormalities in the brain or ventricles, but did exhibit hyperactivity and insomnia phenotypes (Barlow et al., 2023). Finally, overexpression of human ATP1A3 mRNA, or a mutated form associated with RDP and AHC (p.Ala275Pro) resulted in reduced brain size, reduced touch-evoked escape responses, and hypoactivity phenotypes (Ruan et al., 2024). While such inconsistencies likely relate to the methods employed, the relatively mild behavioural phenotypes reported make it unclear if larval zebrafish would recapitulate any of the more dramatic clinical symptoms of ATP1A3-related diseases, such as dystonia and ataxia. Our mutants also appeared morphologically normal, except for deflated swim bladders in some mutant larvae, which is often associated with behavioural abnormalities as inflation requires coordinated swimming to the surface to gulp air. Visual observation in 6-7dpf larvae revealed abnormal swimming patterns and postures, including unbalanced postures with the tail persistently bent to one side, and unbalanced/corkscrew swimming patterns (***Figure 1***C-D). These phenotypes are reminiscent of the abnormal postures and dyscoordination observed in patients with ATP1A3-related disorders, and thus we sought to further characterize these phenotypes genetic associations. Quantifying these effects across X number of larvae revealed that nearly all WT *atp1a3a*$^{+/+}$ larvae displayed normal swimming behavior (96%), while the majority of *atp1a3a*$^{-/-}$ fish were abnormal, displaying either an unbalanced posture (69%) or bent tail (18%) A portion of heterozygous *atp1a3a*$^{+/-}$ larvae were categorized as abnormal (16% unbalanced, 1% bent tail), but the majority were indistinguishable from WT. The distribution of phenotypes in atp1a3a(+/+), atp1a3a(+/-), and atp1a3a(-/-) larvae was significantly different (Fisher-exact test, $p < 0.001$, n=55, 138, 106 for $^{+/+}$, $^{+/-}$ and $^{-/-}$, respectively). These results indicate that the CRISPR induced mutation leads to a variety of abnormal behavioural phenotypes and that there is a degree of haploinsufficiency in the

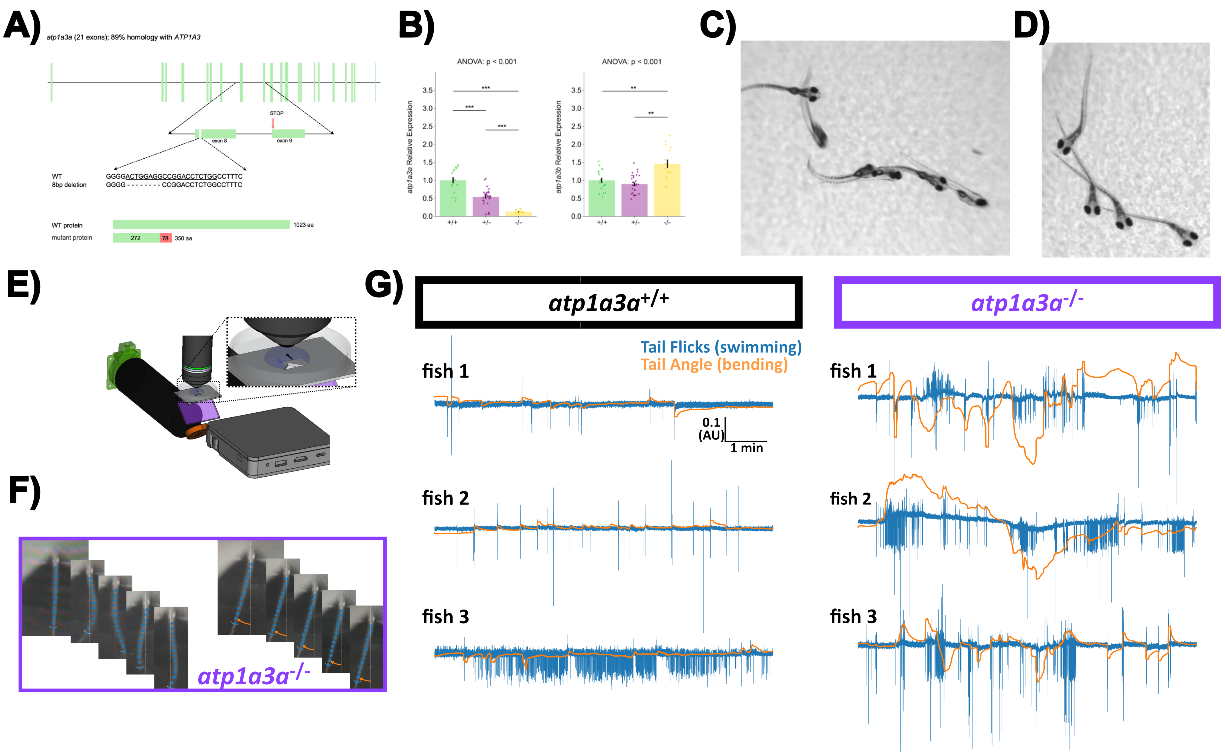

Figure 1. *atp1a3a* mutants display abnormal postures and swimming patterns
A) *atp1a3a* gene structure highlighting the location of the 8 bp deletion in exon 8 that leads to a premature stop codon in exon 9. DNA sequence of target site and predicted atp1a3a protein length in wildtype and mutant zebrafish. gRNA target sequence is underlined. Wildtype atp1a3a consists of 1023 amino acids; atp1a3a mutant protein consists of 272 amino acids identical to wildtype and 78 frameshifted amino acids, yielding a total protein length of 350 amino acids.
B) Comparison of *atp1a3a* and *atp1a3b* expression levels in *atp1a3a* mutants. Expression was quantified using RT-qPCR and normalized to *actb2* expression. Relative expression levels are normalized to expression levels in *atp1a3a*$^{+/+}$ larvae. Data are presented as means ± SEM. * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$.
C) Unbalanced postures and abnormal swimming patterns in *atp1a3a* mutants, including corkscrew swimming, and
D) stable postures with the tail bent to one side.
E) Head-embedded behavioural setup for quantifying tail kinematics in response to visual stimuli. Based on pi_tailtrack (Randlett, 2023)
F) *atp1a3a* mutants display: **i)** Periods of normal swimming interspersed with **ii)** periods with the "bent tail" phenotype with a persistent bend to the left of the midline (dashed line) in this instance.
G) Tail tracking traces of 3 WT larvae showing normal bursts of swimming/tail flicks (blue), with few substantial or long-lasting deviations in the tail angle relative to the midline.
H) Tail tracking traces of 3 *atp1a3a* mutant larvae showing very bursty swimming patterns with vigorous activity followed by periods of quiescence, and many periods of persistent tail bends reflected as large deviations in the Tail Angle.

heterozygous condition, but homozygous mutants show more severe phenotypes, consistent with a loss-of-function mutation in *atp1a3a*.

A key feature of ATP1A3-related disorders is that symptoms often come in waves/attacks, and can be triggered by stress, fever, or exertion (Salles et al., 2021). To see if the phenotypes in *atp1a3a* mutants transient, we made repeated observations/categorizations every hour for 7 hours, as well as once at 7 dpf. This revealed that in a cohort of *atp1a3a*$^{+/-}$ that were categoriezed as "normal" swimmers, 28% transitioned to become abnormal, while 11% became transientlty abnormal before returning to normal. Similarly, from a chorot of *atp1a3a*$^{+/-}$ and *atp1a3a*$^{-/-}$ that were categorized as "abnormal", 20% were subsequently characterized as "normal" in subsequent observations.

To more carefully observe the tail-bending phenotype, we used a head-embedded preparation in which the larva is embedded in agarose with the tail free to move, and presented visual stimuli to elicit swimming behaviour using a projector and screen below the larvae (***Figure 1***E). Observing *atp1a3a* mutants in this setup revealed periods of normal swimming interspersed with periods of persistent tail bends, with the tail often stably bent to one side for many seconds (***Figure 1***F). Tail tracking and swimming kinematic analyses (Randlett, 2023) confirmed that WT sibling fish exhibit the typical bouting swimming pattern, with variable levels of activity in different fish, but that the tail angle remains stably centered along the midline (***Figure 1***E). In contrast, *atp1a3a* mutants show a very bursty swimming pattern, with vigorous bouts of swimming followed by periods of quiescence. Dramatically, these larvae show many periods of persistent tail bends reflected as large deviations in the tail angle from the midline, lasting many seconds or minutes (***Figure 1***F-H).

Based on these data we conclude that the loss of *atp1a3a* results in clear behavioural abnormalities in larval zebrafish, and that these phenotypes appear to often be transient in nature, paralleling this key feature of the human ATP1A3A-related disorders.

atp1a3a mutant phenotypes under stimulus-free conditions

Having established that *atp1a3a* mutants exhibit clear behavioural abnormalities via qualitative observation, we sought to more systematically characterize their phenotypes under spontaneous conditions, and to identify potential underlying neural circuit dysfunctions. For this purpose, we observed larvae in a multi-well setup under normal lighting conditions, in which 300 larvae can be observed simultaneously. (Randlett et al., 2019; Lamiré et al., 2023). Consistent with previous results (Barlow et al., 2023), we found that *atp1a3a*$^{-/-}$ larvae display hyperactivity, characterized by increased total displacement, increased bout frequency, increased median displacement per bout and increased median bout duration (***Figure 2***D-E). Therefore, mutants are hyperactive and show not only initiate more movement bouts, but also move in larger bouts. Under these conditions, we did not observe significant phenotypes in heterozygous *atp1a3a*$^{+/-}$ larvae, indicating that such gross movement phenotypes may be more specific to the homozygous condition, or that the heterozygous condition has more subtle, transient, or less penetrant phenotypes that are not revealed by these population-wide analyses.

In order to identify potential neural substrates and brain areas involved in these abnormal swimming phenotypes, we mapped the brain-wide activity of *atp1a3a* mutants relative to their WT siblings using the MAP-Mapping approach (Randlett et al., 2015). MAP-Mapping uses immunostaining of phospohylated ERK (pERK) relative to total-ERK (tERK) levels as a proxy for neuronal activity, combined with image volume registration and statistical analysis to identify regions with altered pERK/tERK ratios, and presumably altered neuronal activity. We analyzed cohorts of larvae that were let to swim freely in a 10cm petri dish in a brightly lit environment, and determined group genotyping post-imaging. pERK/tERK ratios were compared voxel-wise to generate a brain-wide statistical map of altered activity under these baseline behavioural conditions (***Figure 2***E).

Based on the widespread expression of *atp1a3a* in the brain, as well as the very strong and broad electrophysiological phenotypes expressed in other ATP1A3-deficient models, we expected to see very widespread alterations in neural activity across many brain areas. In contrast, MAP-Mapping revealed a

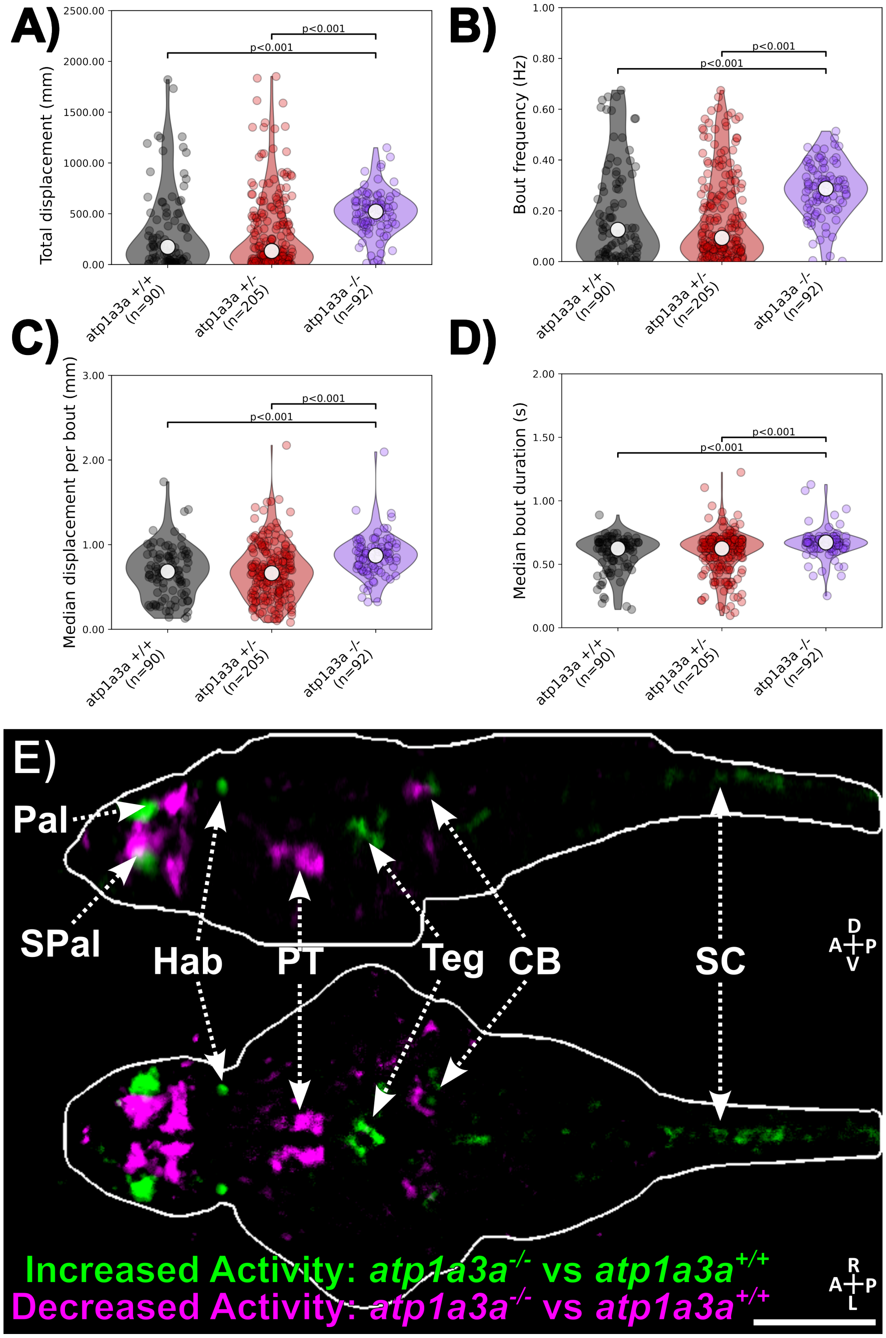

Figure 2. *atp1a3a* mutants are hyperactive and show altered telencephalic activity patterns under spontaneous conditions
Analysis of spontaneous swimming behaviour in 300-well plates (Lamiré et al., 2023) over a 30 min recording period revealed that *atp1a3a* mutants show significantly increased: **A)** total displacement, **B)** movement bout initiation frequency, **C)** displacement per bout, and **D)** duration of bouts. Statistics: Kruskal-Wallis with Dunn's multiple comparisons test.
E) MAP-Mapping analysis of brain-wide activity patterns in *atp1a3a* mutants relative to WT siblings revealed a distributed but fairly restricted pattern of altered activity, with strong activation and suppression patterns in the forebrain, including the pallium (Pal), subpallium (SPal), habenula (Hab), and diencephalon in the posterior tuberculum (PT) and preoptic area. More modest changes are seen in the cerebellum (CB) spinal cord (SC) and midbrain tegmentum (TEG). Scale bar = 100 µm.

distributed but fairly restricted pattern (***Figure 2***E). Consistent with the hyperactivity or perhaps dysregulated motor coordination in these mutants, we saw significant activation in motor-areas, including the spinal cord and midbrain tegmentum, while the cerebellum showed areas of both increased and decreased activity. Strikingly, much more prominent alterations in pERK/tERK signals were seen in the forebrain, including areas, including the pallium, subpallium and habenula. The potential relationships with these altered forebrain activity patterns and hyperactivity is not clear, but do indicate that these areas of the brain appear to be particularly sensitive to *atp1a3a* loss, and thus may be critical for understanding the circuit-level mechanisms underlying the behavioural phenotypes in these mutants.

atp1a3a mutants exhibit altered stimulus-response behaviours

To test the stimulus-response behaviour of *atp1a3a* mutants, we began with our standard 300-well screening assay (***Figure 3***A), in which larvae are exposed to trains of "Dark Flash" stimuli (DF, a sudden transition to darkness that initiates the O-bend response), and acoustic/tap stimuli (a brief vibrational stimulus delivered with a solenoid rod that initiates the C-bend response) (Burgess and Granato, 2007). Stimuli are delivered at 1 minute intervals, and the response probability and kinematics are quantified for each stimulus across the population of larvae. This revealed that *atp1a3a* mutants show a very strongly reduced probability of executing an O-bend response to the DFs (***Figure 3***B). However, visual analyses of the responses to DFs revealed that many mutants do appear to respond to the stimulus, but with a reduced amplitude of bending, and thus may not meet the criteria for a large amplitude O-bend response (***Figure 3***C, Randlett et al., 2019). When we relaxed these criteria to consider any turning response within the 1-second of the DF stimulus, we found that the response probability of mutants was much more similar to siblings (***Figure 3***D), demonstrating that mutants are mostly still able to respond to the stimulus, and only show a significantly reduced probability of responding in the initial phase of the assay (***Figure 3***Ei-iii), but do so with a strongly reduced bending amplitude (***Figure 3***Fi-iii). This may either reflect a reduced ability to execute large amplitude bends, or perhaps that the perceived strength of the stimulus is reduced in mutants, leading to a reduced motor response.

To gain insight into the neural substrates underlying the reduced responses to DFs, we performed MAP-Mapping analysis of brain-wide activity patterns in *atp1a3a* mutants relative to WT siblings after being stimulated with 10 DFs at 1min intervals ()***Figure 3***G). This revealed a very strong suppression of activity in the optic tectum, which is the largest primary visual area in the zebrafish brain, and shows strong responses to DFs in WT larvae (Lamiré et al., 2023). This indicates that the visual DF response is suppressed early in the visual pathway, suggesting sensory deficits early in the circuit.

Suppressed responses were also observed in the cerebellum, and similar to what we observed in the spontaneous condition, we saw relatively subtle alterations in hindbrain areas. Again, the most striking alterations were seen in the forebrain, including the pallium, subpallium and habenula, which show strong patterns of both activation and suppression relative to WT siblings (***Figure 3***G).

In contrast to the reduced responses to DFs, these experiments showed that *atp1a3a* mutants show an increased response probability to acoustic/tap stimuli (***Figure 3***Bi, Di, Eiv), with a bend amplitude indistinguishable from controls (***Figure 3***Fiv). Therefore, *atp1a3a* mutants are not simple hyporesponsive to stimuli, but exhibit very different effects in different assays, indicating a differential role for Atp1a3a in different circuit contexts.

Finally, we

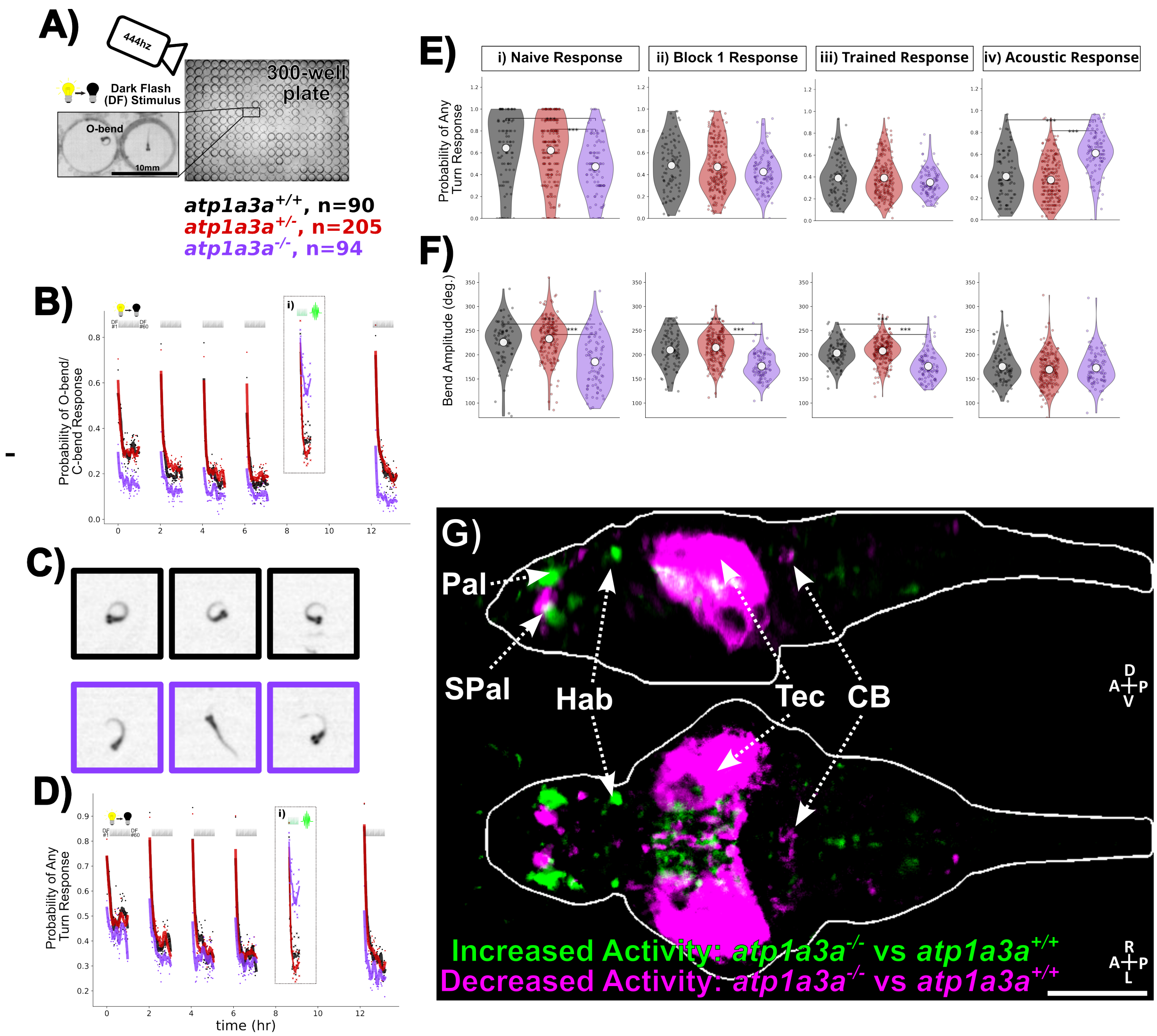

Figure 3. *atp1a3a* mutants show reduced responses to Dark Flash stimuli

A) Multiwell setup for quantifying responses to Dark Flash (DF) stimuli, in which 300 larvae can be observed simultaneously. (Randlett et al., 2019; Lamiré et al., 2023).

B) Response curves comparing *atp1a3a* mutants to heterozygous and WT siblings, showing a reduced response across 5 blocks of 60 DF stimuli delivered at 1 min intervals. Boxed area **(i)**, shows the response to 30 acoustic/"tap" stimuli, delivered at 1min intervals, for which mutants show an exaggerated response relative to siblings. Dots represent mean population response probability for each stimulus, and lines are smoothed across time/stimuli with a Savitzky–Golay filter (window = 15 stimuli, order = 2). O-bend responses to DF and C-bend responses to tap stimuli are defined as turns with a bend amplitude of at least 3 radian, and 1 radian, respectively.

C) Examples of the maximal bending amplitude of three WT (black) and mutant (purple) larvae in response to a dark flash, showing the reduced response amplitude in mutants.

D) Same data as **(B)**, but analyzed to consider any turning response, defined as a bend of at least 1 radian, which reveals a more similar response probability between mutants and siblings, suggesting that mutants are able to respond to the stimulus, but with reduced amplitude.

E) Per-fish quantification of the probability of turning responses, and **F** quantification of the maximal bend amplitude in response to stimuli, for different epochs of the experiment: **i)** the Naive Response to the first 5 DFs, **ii)** the remainder of the first training block (DFs 6-60), **iii)** the trained response in blocks 2-4 (DFs 61-240), and **iv)** the response to tap stimuli (Taps 1-30). Significance tests are Kruskal-Wallis tests with Dunn's multiple comparisons test, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

E) MAP-Mapping analysis of brain-wide activity patterns in *atp1a3a* mutants relative to WT siblings after being stimulated with 10 DFs at 1min intervals. The signals in the optic tectum (Tec) are stronlgy suppressed, while the cerebellum (CB) and hindbrain areas show more modest alterations. Strong activation and suppression patterns are seen in the forebrain, including the pallium (Pal), subpallium (SPal), habenula (Hab). Scale bar = 100 µm.

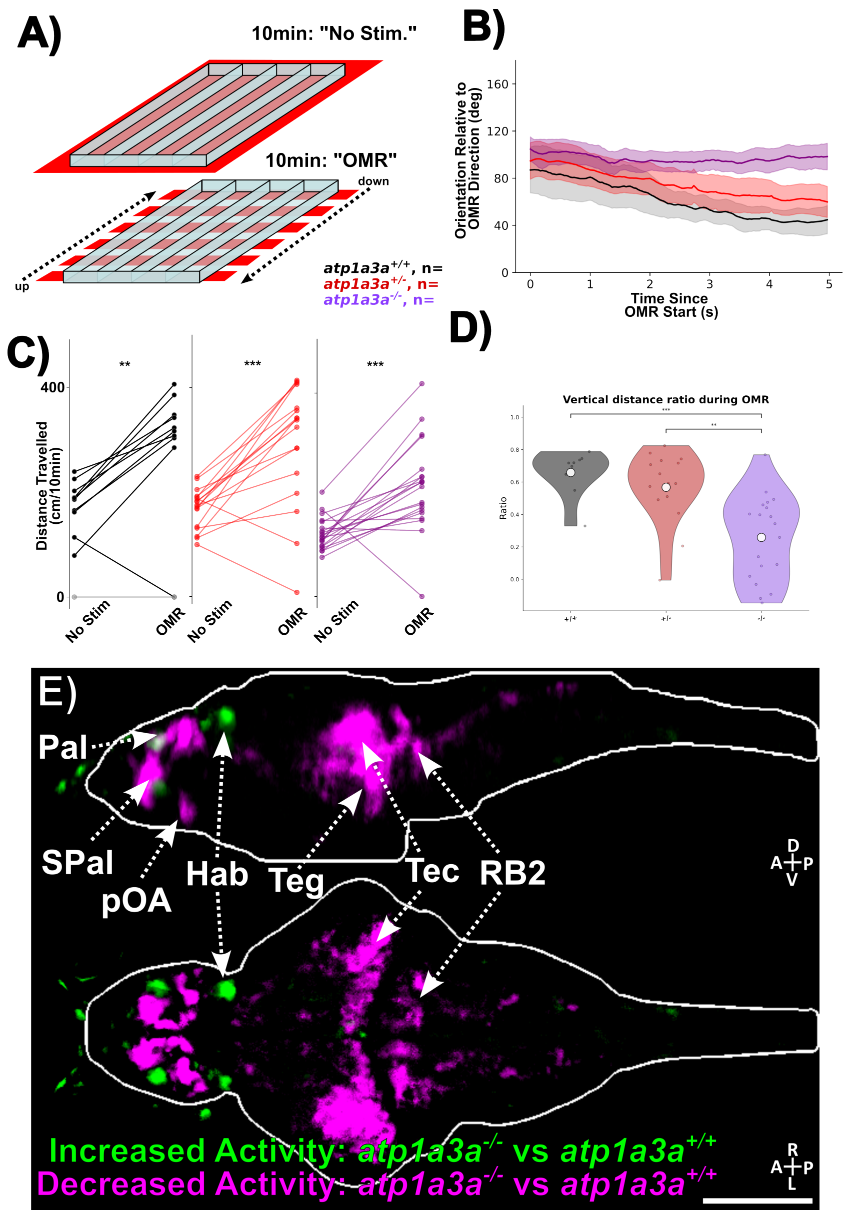

Figure 4. *atp1a3a* mutants show reduced responses directional response to Optomotor stimuli
A) Arena for Optomotor Response (OMR) assay, in which larvae are placed in tracks and either presented with global red light for the stimulus-free condition ("No Stim."), or with moving striped patterns that elicit robust directional swimming responses in WT larvae ("OMR Stim.").

Materials and Methods

Animal husbandry and care

All experiments were carried out using larval zebrafish between 5 and 6 days post-fertilization (dpf), maintained in E3 embryo medium supplemented with 0.02Larvae were housed in 10 cm petri dishes under a 14:10 h light/dark cycle at a constant temperature of 28-29°C. Adult zebrafish were maintained in conditioned system water at 28°C under identical light/dark conditions, and fed twice daily with live Artemia (brine shrimp). Zebrafish were bred and maintained at the SickKids Zebrafish CORE facility (Toronto, Canada) and the Animalerie Zebrafish Rockefeller (AZR, SFR Santé Lyon Est, Lyon). All housing and experimental procedures were conducted in accordance with institutional and national regulations. French facilities were approved by the French Ministry of Agriculture and the Ministry of Higher Education and Research (Agreement #C693870602) and overseen by the local ethics committee (comité d'éthique en expérimentation animale de la Région Rhône-Alpes: CECCAPP). Experiments conducted in Canada were approved by the University of Toronto Animal Care Committee, in accordance with guidelines established by the Canadian Council for Animal Care (CCAC).

Generation of the *atp1a3a* mutant line

The *atp1a3a* CRISPR mutant zebrafish line was generated at the SickKids Zebrafish CORE facility (Toronto, Canada). The guide RNA (gRNA) was designed to target the *atp1a3a* gene 82 base pairs downstream of the start of exon 8. This specific region was chosen due to its close proximity to two human mutations known to cause Rapid-onset Dystonia Parkinsonism (821T>C mutation and 829G>A mutation). The resultant *atp1a3a* mutant allele has an 8 base pair deletion in exon 8 leading to a frameshift mutation with a premature stop in exon 9. For Ca^{2+} imaging experiments, *atp1a3a* mutants were crossed with *Tg(elavl3:Huc-GCaMP6f); mitfa*$_{-/-}$, to generate a line that express the nuclear-targeted GCaMP6f Ca2+ indicator pan-neuronally, in the pigment deficient *Nacre* mutants.

Genotyping

Larvae were lysed in 50 mM NaOH at 95°C for 10 min and neutralized with 1 M Tris-HCl (pH 8.0) in a 1:9 volume ratio (Tris:NaOH), to extract genomic DNA. Target region of atp1a3a was amplified using FIREPol® Master Mix Ready to Load 1× (Solis Biodyne) and gene-specific primers at 0.1 µM. Primer sequences used were: Forward : CGGTATTGTGGTCTGCACCG, Reverse: GCCGGTGATGATGTGGATGA. Wild-type alleles yielded amplicons of 119 bp, while mutant alleles are 111 bp, which were distinguished on an 3.5% agarose gel.

Analyses of Spontaneous Behaviour

The spontaneous behaviour of larvae was recorded using a high-speed imaging system adapted from BonZeb-based kinematic tracking (Guilbeault et al., 2021). The setup included a Sentech STC-CMB200PCL CameraLink camera (Aegis Electronic Group, Inc.) with a zoom lens (Kowa), connected via a GrabLink Full XR Frame Grabber (Euresys Inc.) for acquisition of 1088 × 1088 resolution frames at 332 Hz. Larvae were placed on a 6 mm thick transparent polycarbonate platform, illuminated from below by an 850 nm LED plate. A 1920 × 1080, 60 Hz DLP pico projector (P7, AAXA Technologies Inc.) was integrated into the system, with projected light directed onto the platform using a cold mirror (Knight Optical). Real-time video acquisition and processing were carried out in the Bonsai programming environment (Lopes et al., 2015), which interfaced with BonZeb for automated tracking (Guilbeault et al., 2021). This configuration was used to obtain free-swimming recordings of *atp1a3a* mutants, from which subsets of frames were extracted in FIJI and z-projected to generate composite images illustrating abnormal phenotypes and swim sequences. Free swimming was further analyzed at 5 dpf using the 300-well plate setup we have previously described, where larvae are recorded in 8mm x 6mm (diameter x depth) wells (Lamiré et al., 2023). Larvae were recorded over a 30-minute period in the absence of visual and acoustic stimulation. From these recordings, the mean displacement per bout (mm), mean bout duration (s), mean peak bout speed (mm/s), total displacement (mm), and bout frequency (Hz) were calculated using custom python scripts (available at github.com/owenrandlett/2025_atp1a3a).

Analyses of the Optomotor Response (OMR)

The OMR was assessed in 5dpf larvae in an acrylic arena containing 3 separate corridors allowing fish to swim in a 10.4cm x 2cm arena. Larvae were subjected to alternating vertically moving OMR gratings using a projector (AAXA LED Pico Projector) and a 45° cold mirror. OMR gratings were calcualted and dynamically displpayed using the "PsychoPy" package in python (Peirce, 2007) The gratings ran continually for 20 seconds epochs, alternating between directions along the long-axis of the arena, spanned by a 10 second long "no-stimulus" interval. In total, the alternating OMR stimuli ran for 10 minutes, subjecting each larva to a total of 20 repeats of the stimuli. Fish were tracked using IR LED arrays, and a camera (Chameleon 3 USB3 FLIR) running at 30 frames per second with an IR filter, and custom machine-vision scripts (available at github.com/owenrandlett/2025_atp1a3a). Lighting was projected at a screen under the arena using a projector (AAXA LED Pico Projector) and a 45° cold mirror.

Analyses of the Dark Flash Response

Larval behavior was assessed in custom 300-well acrylic plates (8 mm diameter, 6 mm depth; 300 μL water volume) based on a modified version of previously described setups.

Larvae from adult atp1a3a+/-; Tg(elavl3:Huc-GCaMP6f) in-crosses were used to track behaviour [258, 273].

The plates were positioned beneath a 31°C water bath, which served as a heated lid to prevent condensation and maintain the wells at 29°C.

Recordings were captured using a Mikrotron CXP-4 camera (500 Hz) connected to a Silicon Software frame grabber (Marathon ACX-QP, Basler) and illuminated with IR LEDs (TSHF5410, digikey.com).

Visual stimulation was provided by a rectangular array of 155 WS2813 RGB LEDs, with dark flashes (DFs) consisting of a 1 second light-off period followed by a linear return to baseline over 20 seconds.

Acoustic/vibration taps were delivered via a solenoid (ROB-10391, Sparkfun).

The experimental design alternated 1 hour of stimulation with 1 hour of rest, during which a stepper motor-driven actuator (Hanpose HPV4, 500 cm) moved the camera between two plates, enabling up to 600 larvae to be tested simultaneously.

Control of the LEDs, solenoid, and camera actuator was managed by a Raspberry Pi Pico running CircuitPython and custom Python software, which also handled video acquisition and online tracking of head and tail positions at 20-30 Hz.

During DF or tap delivery, a 1 second high-speed burst recording was captured and later analyzed offline using background subtraction and morphological operations.

Tracking and analysis relied on open-source libraries, including OpenCV, scikit-image, NumPy, SciPy, and Numba.

MAP-Mapping Analyses of Brain-Wide Activity Patterns

Whole-brain activity mapping was performed using a MAP-mapping pipeline adapted from Randlett et al. (2015).

Cohorts of 150-200 larvae produced from adult atp1a3a+/- ; mitfa-/- ; Tg(elavl3:Huc-GCaMP6f) incross were subjected to visual stimuli as following:

For free swimming, larvae were grouped in one 10 cm petri dish and left swimming in a lit environment for 40 minutes in the same behavior setup as was used for the OMR stimulation (described below) to ensure identical external conditions.

For OMR stimulation, larvae were grouped in one 10 cm petri dish, left to acclimatize for 10 minutes then subjected to concentric gratings that alternated vertically each 30 seconds, using the OMR behavior setup.

For dark flash stimulation, larvae were grouped in one 10 cm petri dish, left to acclimatize for 20 minutes then subjected to 10 dark flashes following the dark flash protocol and using the dark flash habituation setup.

Larvae were then fixed and processed for immunohistochemistry with anti-pERK (Cell Signaling Technology, #4370S) and anti-tERK primary antibodies (Cell Signaling Technology, #4696S), followed by fluorophore-conjugated secondary antibodies: Alexa FluorTM 647 for tERK (Thermo Fisher Scientific, #A21235) and Alexa FluorTM 568 for pERK (Thermo Fisher Scientific, #11036).

Brains were imaged in dorsal orientation on the Leica Stellaris 5 confocal microscope equipped with a 25× water-immersion objective (HC FLUOTAR L 25x/0.95 W VISIR, CIQLE imaging platform, Lyon).

Image stacks were acquired at a voxel resolution of 0.08 × 0.08 × 2 μm, and two partially overlapping volumes were tiled to encompass the entire brain.

The GCaMP6f channel was used as an anatomical reference for nonlinear registration of each specimen to a standardized reference brain via the Computational Morphometry Toolkit (CMTK).

Registered image volumes were subsequently down-sampled to 300 × 679 × 80 voxels (x, y, z) and spatially smoothed using a two-dimensional Gaussian filter.

To account for inter-individual variability in staining intensity and morphology, voxel-wise pERK signals were normalized to corresponding tERK values prior to statistical comparison across experimental conditions.

Calcium imaging

Data analysis

For the initial characterisation performed by Shahrzad, phenotypic distributions were visualized and analyzed using multiple statistical approaches.

Pie charts depicting the relative proportions of phenotypes in atp1a3a mutants were generated with Python packages Matplotlib, Pandas, and NumPy.

The same visualization pipeline was employed for phenotype transience assays, while histograms of tail angle distributions were constructed using these packages as well.

Statistical evaluation of swim kinematics during OMR assays was conducted with the Pingouin library.

Outliers were identified as values exceeding 1.5 times the interquartile range below the first quartile or above the third quartile and were excluded prior to subsequent analyses.

For RT-qPCR, relative expression levels of atp1a3a and atp1a3b were quantified using the $\Delta\Delta Ct$ method.

Data were analyzed with one-way ANOVAs followed by Bonferroni-corrected post hoc t-tests.

Outlier replicates were defined as those with $|z| > 3.5$ and removed prior to analysis.

A threshold of $p < 0.05$ was adopted to determine statistical significance in all tests.

Data analysis was carried out in Python using a custom-developed script.

Behavioral events were operationally defined based on cumulative tail bend angle thresholds, with deflections greater than 3 radians classified as O-bends and those exceeding 1 radian designated as C-bends.

The computational framework integrated several open-source libraries: NumPy, Pandas, Numba and SciPy for statistical procedures.

Data visualization was performed using Matplotlib and seaborn.

Statistical comparisons between distributions were conducted using the non-parametric Mann-Whitney U test as implemented in SciPy.

Discussion

The promotion of habituation learning by estradiol is mediated by an unidentified target

Our experiments indicate that Gper1, ER1, ER2a and ER2b do not mediate the positive effects of estradiol on habituation learning. As this is fundamentally a negative result, it is difficult to conclusively demonstrate this beyond any doubt. One major caveat relates to the actual functional nature of the mutant alleles that we have used. These are all Cas9-generated small deletions resulting in frameshift mutations that lead to early stop codons, and are thus predicted null/knockout lines. Despite this genetic confidence, it is always possible that residual activity could still remain, perhaps via alternate splicing or alternate start codons. This could be further complicated by genetic/transcriptional compensation, where frameshift alleles can lead to the upregulation of paralogs in some circumstances (El-Brolosy et al., 2019). As with all negative results, it is not possible to rule out all possible alternative explanations. However, we recognize that this bias against publishing negative results is bad for science. Negative results from well-designed and executed experiments are of value for the community and making this knowledge public is our duty as responsible scientists (Mlinarić et al., 2017).

While the possibility of "residual activity" in our mutants is a clear limitation of our approach, we argue that this alternative interpretation is very unlikely. The *esr1^{uab118}* and *esr2b^{uab127}* alleles both exhibited a lack of estradiol responsiveness in other tissues (Romano et al., 2017), and *esr2b^{uab127}* mutants are female sterile/subfertile (D. Gorelick, personal communication), indicating a non-functional receptor. Similarly, *gper1^{uab102}* mutants show a lack of estradiol responsiveness in heart rate modulation (Romano et al., 2017). Interestingly this was only observed in maternal-zygotic mutants. While it seems unlikely that sufficient maternal mRNA/protein for Gper1 could persist in 5dpf larvae, we can formally rule this out with our current datasets. The *esr2a^{uab134}* mutants have no previously published phenotype, and so we do not have an independent positive control for the nature of this allele. However, the best evidence we have against the "residual activity" hypothesis is that we actually found phenotypes in our assays for *esr1^{uab118}*, *esr2a^{uab134}*, and *gper1^{uab102}* mutants. These phenotypes are just of the unexpected sign, where mutants show increased habituation (discussed below).

Candidate estradiol targets that could promote habituation learning

We have concluded that the lack of habituation deficits in our mutants is due to the presence of an alternative receptor or pathway that mediates the learning-promoting effects of estradiol. What might this unidentified target be? Various leads exist in the literature. One hypothesis posits the existence of an unidentified “Gq-mER” (Gq-coupled membrane estrogen receptor) (Qiu et al., 2006; Vail and Roepke, 2019), and therefore estradiol may signal via additional GPCRs beyond Gper1. Another possibility is an interaction between estradiol and other membrane receptors, for example: the Voltage-Gated Sodium Channel Nav1.2 (Sula et al., 2021; Treviño and Gorelick, 2021), transient receptor potential (TRP) channels (Payrits et al., 2017; Ramírez-Barrantes et al., 2020), or various other ion channels (Kow and Pfaff, 2016). The robust nature of our "non-canonical" but clearly estradiol-dependent phenotype, combined with the high-throughput nature of our behavioural assays, could be an ideal assay for future screening efforts to attempt to identify novel estradiol target(s).

Multiple Estrogen Receptors act to suppress habituation learning.

While we were surprised to find that the classical ERs do not promote habituation, we were shocked to find evidence of the opposite! We found that *esr1^{uab118}*, *esr2a^{uab134}*, and *gper1^{uab102}* mutants habituate more than their sibling controls (***Figure 1***, ***Figure 2***), consistent with a role for these ERs in inhibiting habituation. While effect sizes of these magnitudes border on those that are easily dismissible as "noise", they were not only observed in the single mutants experiments, but also in the double and triple mutant combinations of these alleles (***Figure 3***-**??**), providing good evidence that they are biologically meaningful effects. In fact, these multi-mutants generally exhibited larger effect sizes, consistent with an additive interaction.

Untangling the mechanisms of ER1-, ER2a- and Gper1-dependent suppression of habituation will require considerable further work. The additive interaction we observed genetically indicates that the ERs act cooperatively to suppress habituation learning. All three receptors are expressed in the larval zebrafish brain (Thisse and Thisse, 2008; Romano et al., 2017), but whether they are acting in the same or different cell types awaits characterization. The study of this inhibitory pathway may be challenging since it opposes the major learning-promoting effect of estradiol, and therefore may be more straightforward to study after the identification and deletion of the estradiol target that promotes habituation. One attractive hypothesis relates to the observation that estradiol exposure increases aromatase expression in the zebrafish brain (Menuet et al., 2005; Hao et al., 2013). Since aromatase catalyzes the conversion of androgens into estrogens, ER mutants may have reduced levels of endogenous endogeous estrogens in the brain, which could lead to inhibited habituation. Future experiments aimed at manipulating the aromatase system in wild-type and ER mutant backgrounds could be used to test this hypothesis.

Conclusion

What began as a straightforward study to identify the receptor(s) that mediate the habituation-promoting effects of estradiol has instead led us to a surprising and paradoxical result; canonical ERs do regulate habituation, but are suppressive and act in opposition to the habituation-promoting effects of estradiol. This fits with the general theme of our studies of this habituation paradigm – we find increasing complexity and contradiction within this "simple" learning process the deeper we look. This began with our detailed observations of behaviour, leading us to conclude that habituation results from a distributed plasticity process that adapts different aspects of behavior independently (Randlett et al., 2019). We believe that this property underlies our subsequent discoveries of pharmacological and genetic manipulations that can result in either specific changes in specific aspects of habituation (but not others), or even opposing effects, where a single manipulation can simultaneously increase and decrease habituation, depending on which component of behavior is measured (Randlett et al., 2019; Lamiré et al., 2023). This complexity appears to be a fundamental property of habituation (McDiarmid et al., 2019), and that the study of habituation will likely continue to surprise us, hopefully leading to unexpected insights into the nature of plasticity underlying learning and memory.

Data Availability Statement

Original data generated and analyzed during this study are included in this published article or in the data repositories listed in References.

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References

F Alarcón, E Tolosa, and E Muñoz. Focal limb dystonia in a patient with a cerebellar mass. *Arch. Neurol.*, 58(7):1125–1127, July 2001.

Alberto Albanese, Kailash Bhatia, Susan B Bressman, Mahlon R Delong, Stanley Fahn, Victor S C Fung, Mark Hallett, Joseph Jankovic, Hyder A Jinnah, Christine Klein, Anthony E Lang, Jonathan W Mink, and Jan K Teller. Phenomenology and classification of dystonia: a consensus update. *Mov. Disord.*, 28(7):863–873, June 2013.

Maurizio Balestrino, Jacob Young, and Peter Aitken. Block of (na+, k+)atpase with ouabain induces spreading depression-like depolarization in hippocampal slices. *Brain Research*, 838(1–2):37–44, August 1999. ISSN 0006-8993. doi: 10.1016/s0006-8993(99)01674-1. URL http://dx.doi.org/10.1016/s0006-8993(99)01674-1.

Ida L Barlow, Eirinn Mackay, Emily Wheater, Aimee Goel, Sumi Lim, Steve Zimmerman, Ian Woods, David A Prober, and Jason Rihel. The zebrafish mutant dreammist implicates sodium homeostasis in sleep regulation. *eLife*, 12, August 2023. ISSN 2050-084X. doi: 10.7554/elife.87521. URL http://dx.doi.org/10.7554/eLife.87521.

Harold A. Burgess and Michael Granato. Modulation of locomotor activity in larval zebrafish during light adaptation. *Journal of Experimental Biology*, 210(14):2526–2539, July 2007. ISSN 0022-0949. doi: 10.1242/jeb.003939. URL http://dx.doi.org/10.1242/jeb.003939.

D Paola Calderon, Rachel Fremont, Franca Kraenzlin, and Kamran Khodakhah. The neural substrates of rapid-onset Dystonia-Parkinsonism. *Nat. Neurosci.*, 14(3):357–365, March 2011.

Canan Doğanlı, Hans C. Beck, Angeles B. Ribera, Claus Oxvig, and Karin Lykke-Hartmann. α3na+/k+-atpase deficiency causes brain ventricle dilation and abrupt embryonic motility in zebrafish. *Journal of Biological Chemistry*, 288(13):8862–8874, March 2013. ISSN 0021-9258. doi: 10.1074/jbc.m112.421529. URL http://dx.doi.org/10.1074/jbc.M112.421529.

Jens P Dreier. The role of spreading depression, spreading depolarization and spreading ischemia in neurological disease. *Nature Medicine*, 17(4):439–447, April 2011. ISSN 1546-170X. doi: 10.1038/nm.2333. URL http://dx.doi.org/10.1038/nm.2333.

Mohamed A El-Brolosy, Zacharias Kontarakis, Andrea Rossi, Carsten Kuenne, Stefan Günther, Nana Fukuda, Khrievono Kikhi, Giulia L M Boezio, Carter M Takacs, Shih-Lei Lai, Ryuichi Fukuda, Claudia Gerri, Antonio J Giraldez, and Didier Y R Stainier. Genetic compensation triggered by mutant mRNA degradation. *Nature*, 568(7751):193–197, April 2019.

Nicholas C. Guilbeault, Jordan Guerguiev, Michael Martin, Isabelle Tate, and Tod R. Thiele. Bonzeb: open-source, modular software tools for high-resolution zebrafish tracking and analysis. *Scientific Reports*, 11(1), April 2021. ISSN 2045-2322. doi: 10.1038/s41598-021-85896-x. URL http://dx.doi.org/10.1038/s41598-021-85896-x.

Ruixin Hao, Maria Bondesson, Amar V Singh, Anne Riu, Catherine W McCollum, Thomas B Knudsen, Daniel A Gorelick, and Jan-Åke Gustafsson. Identification of estrogen target genes during zebrafish embryonic development through transcriptomic analysis. *PLoS One*, 8(11):e79020, November 2013.

Ihtsham U Haq, Beverly M Snively, Kathleen J Sweadner, Cynthia K Suerken, Jared F Cook, Laurie J Ozelius, Charlotte Miller, William V McCall, Christopher T Whitlow, and Allison Brashear. Revising rapid-onset dystonia-parkinsonism: Broadening indications for ATP1A3 testing. *Mov. Disord.*, 34(10):1528–1536, October 2019.

Thomas H Holm and Karin Lykke-Hartmann. Insights into the pathology of the α3 Na(+)/K(+)-ATPase ion pump in neurological disorders; lessons from animal models. *Front. Physiol.*, 7:209, June 2016.

Arsen S. Hunanyan, Nina A. Fainberg, Molly Linabarger, Eric Arehart, A. Soren Leonard, Syed M. Adil, Ashley R. Helseth, Amanda K. Swearingen, Stacy L. Forbes, Ramona M. Rodriguiz, Theodore Rhodes, Xiaodi Yao, Nadine Kibbi, Daryl W. Hochman, William C. Wetsel, Ute Hochgeschwender, and Mohamad A. Mikati. Knock-in mouse model of alternating hemiplegia of childhood: Behavioral and electrophysiologic characterization. *Epilepsia*, 56(1):82–93, December 2014. ISSN 1528-1167. doi: 10.1111/epi.12878. URL http://dx.doi.org/10.1111/epi.12878.

Lee-Ming Kow and Donald W Pfaff. Rapid estrogen actions on ion channels: A survey in search for mechanisms. *Steroids*, 111:46–53, July 2016.

Sheng-Han Kuo. Ataxia. *Continuum (Minneap. Minn.)*, 25(4):1036–1054, August 2019.

Laurie Anne Lamiré, Martin Haesemeyer, Florian Engert, Michael Granato, and Owen Randlett. Functional and pharmacological analyses of visual habituation learning in larval zebrafish. *Elife*, 12, December 2023.

Yinchao Li, Xianyue Liu, Chengzhe Wang, Zhengwei Su, Ke Zhao, Man Yang, Shuda Chen, and Liemin Zhou. Molecular and clinical characteristics of ATP1A3-related diseases. *Front. Neurol.*, 13:924788, July 2022.

GonÃ§alo Lopes, NiccolÃ² Bonacchi, JoÃ£o FrazÃ£o, Joana P. Neto, Bassam V. Atallah, Sofia Soares, LuÃs Moreira, Sara Matias, Pavel M. Itskov, PatrÃcia A. Correia, Roberto E. Medina, Lorenza Calcaterra, Elena Dreosti, Joseph J. Paton, and Adam R. Kampff. Bonsai: an event-based framework for processing and controlling data streams. *Frontiers in Neuroinformatics*, 9, April 2015. ISSN 1662-5196. doi: 10.3389/fninf.2015.00007. URL http://dx.doi.org/10.3389/fninf.2015.00007.

Troy A McDiarmid, Alex J Yu, and Catharine H Rankin. Habituation is more than learning to ignore: Multiple mechanisms serve to facilitate shifts in behavioral strategy. *Bioessays*, 41(9):e1900077, September 2019.

Arnaud Menuet, Elisabeth Pellegrini, François Brion, Marie-Madeleine Gueguen, Isabelle Anglade, Farzad Pakdel, and Olivier Kah. Expression and estrogen-dependent regulation of the zebrafish brain aromatase gene. *J. Comp. Neurol.*, 485(4): 304–320, May 2005.

Jonathan W Mink. The basal ganglia and involuntary movements: impaired inhibition of competing motor patterns. *Arch. Neurol.*, 60(10):1365–1368, October 2003.

Jonathan W Mink. Basal ganglia mechanisms in action selection, plasticity, and dystonia. *Eur. J. Paediatr. Neurol.*, 22(2): 225–229, March 2018.

Ana Mlinarić, Martina Horvat, and Vesna Šupak Smolčić. Dealing with the positive publication bias: Why you should really publish your negative results. *Biochem. Med. (Zagreb)*, 27(3):030201, October 2017.

Maja Payrits, Éva Sághy, Kata Csekő, Krisztina Pohóczky, Kata Bölcskei, Dávid Ernszt, Klaudia Barabás, János Szolcsányi, István M Ábrahám, Zsuzsanna Helyes, and Éva Szőke. Estradiol sensitizes the transient receptor potential vanilloid 1 receptor in pain responses. *Endocrinology*, 158(10):3249–3258, October 2017.

Jonathan W. Peirce. Psychopy—psychophysics software in python. *Journal of Neuroscience Methods*, 162(1–2):8–13, May 2007. ISSN 0165-0270. doi: 10.1016/j.jneumeth.2006.11.017. URL http://dx.doi.org/10.1016/j.jneumeth.2006.11.017.

S J Pittock, C Joyce, V O'Keane, B Hugle, M O Hardiman, F Brett, A J Green, D E Barton, M D King, and D W Webb. Rapid-onset dystonia-parkinsonism: a clinical and genetic analysis of a new kindred. *Neurology*, 55(7):991–995, October 2000.

Jian Qiu, Martha A Bosch, Sandra C Tobias, Andree Krust, Sharon M Graham, Stephanie J Murphy, Kenneth S Korach, Pierre Chambon, Thomas S Scanlan, Oline K Rønnekleiv, and Martin J Kelly. A g-protein-coupled estrogen receptor is involved in hypothalamic control of energy homeostasis. *J. Neurosci.*, 26(21):5649–5655, May 2006.

Ricardo Ramírez-Barrantes, Karina Carvajal-Zamorano, Belen Rodriguez, Claudio Cordova, Carlo Lozano, Felipe Simon, Paula Díaz, Pablo Muñoz, Ivanny Marchant, Ramón Latorre, Karen Castillo, and Pablo Olivero. TRPV1-estradiol stereospecific relationship underlies cell survival in oxidative cell death. *Front. Physiol.*, 11:444, May 2020.

Owen Randlett. pi_tailtrack: A compact, inexpensive and open-source behaviour-tracking system for head-restrained zebrafish. *Journal of Experimental Biology*, 226(22), November 2023. ISSN 1477-9145. doi: 10.1242/jeb.246335. URL http://dx.doi.org/10.1242/jeb.246335.

Owen Randlett, Caroline L Wee, Eva A Naumann, Onyeka Nnaemeka, David Schoppik, James E Fitzgerald, Ruben Portugues, Alix M B Lacoste, Clemens Riegler, Florian Engert, and Alexander F Schier. Whole-brain activity mapping onto a zebrafish brain atlas. *Nature Methods*, 12(11):1039–1046, September 2015. ISSN 1548-7105. doi: 10.1038/nmeth.3581. URL http://dx.doi.org/10.1038/nmeth.3581.

Owen Randlett, Martin Haesemeyer, Greg Forkin, Hannah Shoenhard, Alexander F Schier, Florian Engert, and Michael Granato. Distributed plasticity drives visual habituation learning in larval zebrafish. *Curr. Biol.*, 29(8):1337–1345.e4, April 2019.

Shannon N Romano, Hailey E Edwards, Jaclyn Paige Souder, Kevin J Ryan, Xiangqin Cui, and Daniel A Gorelick. G protein-coupled estrogen receptor regulates embryonic heart rate in zebrafish. *PLoS Genet.*, 13(10):e1007069, October 2017.

Andrea Rossi, Zacharias Kontarakis, Claudia Gerri, Hendrik Nolte, Soraya Hölper, Marcus Krüger, and Didier Y. R. Stainier. Genetic compensation induced by deleterious mutations but not gene knockdowns. *Nature*, 524(7564):230–233, July 2015. ISSN 1476-4687. doi: 10.1038/nature14580. URL http://dx.doi.org/10.1038/nature14580.

Dan-dan Ruan, Jing Zou, Li-sheng Liao, Ming-dong Ji, Ruo-li Wang, Jian-hui Zhang, Li Zhang, Mei-zhu Gao, Qian Chen, Hong-ping Yu, Wen Wei, Yun-fei Li, Hong Li, Fan Lin, Jie-wei Luo, and Xin-fu Lin. In vitro study of atp1a3 p.ala275pro mutant causing alternating hemiplegia of childhood and rapid-onset dystonia-parkinsonism. *Frontiers in Neuroscience*, 18, July 2024. ISSN 1662-453X. doi: 10.3389/fnins.2024.1415576. URL http://dx.doi.org/10.3389/fnins.2024.1415576.

Philippe A Salles, Ignacio F Mata, Tobias Brünger, Dennis Lal, and Hubert H Fernandez. ATP1A3-related disorders: An ever-expanding clinical spectrum. *Front. Neurol.*, 12:637890, April 2021.

Altin Sula, David Hollingworth, Leo C T Ng, Megan Larmore, Paul G DeCaen, and B A Wallace. A tamoxifen receptor within a voltage-gated sodium channel. *Mol. Cell*, 81(6):1160–1169.e5, March 2021.

Matthew T Sweney, Kenneth Silver, Marion Gerard-Blanluet, Jean-Michel Pedespan, Francis Renault, Alexis Arzimanoglou, Mylynda Schlesinger-Massart, Aga J Lewelt, Sandra P Reyna, and Kathryn J Swoboda. Alternating hemiplegia of childhood: early characteristics and evolution of a neurodevelopmental syndrome. *Pediatrics*, 123(3):e534–41, March 2009.

Christine Thisse and Bernard Thisse. High-resolution in situ hybridization to whole-mount zebrafish embryos. *Nat. Protoc.*, 3(1):59–69, 2008.

Lindsey S Treviño and Daniel A Gorelick. The interface of nuclear and membrane steroid signaling. *Endocrinology*, 162(8), August 2021.

Gwyndolin Vail and Troy A Roepke. Membrane-initiated estrogen signaling via gq-coupled GPCR in the central nervous system. *Steroids*, 142:77–83, February 2019.

Aikaterini Vezyroglou, Rhoda Akilapa, Katy Barwick, Saskia Koene, Catherine A Brownstein, Muriel Holder-Espinasse, Andrew E Fry, Andrea Hilary Nemeth, George K Tofaris, Eleanor Hay, Imelda Hughes, Sahar Mansour, Santosh R Mordekar, Miranda Splitt, Peter D Turnpenny, Demetria Demetriou, Tamara T Koopmann, Claudia A L Ruivenkamp, Pankaj B Agrawal, Lucinda Carr, Virginia Clowes, Neeti Ghali, Susan Elizabeth Holder, Jessica Radley, Alison Male, Sanjay M Sisodiya, Manju A Kurian, J Helen Cross, and Meena Balasubramanian. The phenotypic continuum of ATPLA3-related disorders. *Neurology*, 99(14):e1511–e1526, July 2022.

An Vo, Nha Nguyen, Koji Fujita, Katharina A Schindlbeck, Andrea Rommal, Susan B Bressman, Martin Niethammer, and David Eidelberg. Disordered network structure and function in dystonia: pathological connectivity vs. adaptive responses. *Cereb. Cortex*, 33(11):6943–6958, May 2023.